

ARTYOMOVSK, Russian Federation

WMO: 297720

Lat: 54.3475N

Lon: 93.4350E

Elev: 479

StdP: 95.70

Time Zone: 7.00 (E07)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.1	-30.8	-36.8	0.1	-33.9	-33.4	0.2	-30.7	6.8	-8.1	5.4	-11.0	0.1	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	29.5	18.3	27.5	17.5	25.5	16.8	19.7	26.3	18.8	24.8	17.9	23.3	1.8	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.6	13.4	21.9	16.7	12.6	20.8	16.0	12.0	20.0	58.8	26.6	55.6	24.7	52.6	23.2	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.3	5.2	4.3	-36.9	32.3	3.8	2.3	-39.6	33.9	-41.8	35.3	-44.0	36.6	-46.7	38.3		
(5)			-37.3	21.9	3.8	0.8	-40.0	22.5	-42.3	23.0	-44.4	23.4	-47.2	24.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.8	-17.1	-13.4	-5.1	2.3	8.0	15.3	17.9	15.7	8.7	0.9	-10.0	-14.7		
	DBStd	13.15	6.62	6.20	5.69	4.45	4.24	4.48	2.73	3.12	3.99	4.77	6.95	6.95		
	HDD10.0	4014	841	654	467	232	92	9	0	1	72	282	599	767		
	HDD18.3	6464	1099	888	726	481	322	111	41	94	290	539	849	1025		
	CDD10.0	649	0	0	0	2	28	167	244	176	32	1	0	0		
	CDD18.3	57	0	0	0	0	1	19	27	11	0	0	0	0		
	CDH23.3	931	0	0	0	1	43	349	341	178	18	0	0	0		
	CDH26.7	245	0	0	0	0	8	102	93	40	2	0	0	0		
(14) Wind	WSAvg	1.4	0.7	1.2	1.5	1.9	2.0	1.6	1.5	1.4	1.3	1.4	1.1	0.7		
(15) Precipitation	PrecAvg	760	29	25	34	50	67	86	106	109	77	66	65	47		
	PrecMax	920	48	75	77	84	116	128	180	189	140	193	112	95		
	PrecMin	602	9	8	10	21	18	42	52	64	26	21	38	13		
	PrecStd	80	10	13	17	18	23	24	25	26	27	31	18	19		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-1.7	2.7	12.6	19.7	27.7	31.9	31.7	30.3	25.9	17.9	6.1	0.9		
		MCWB	-2.3	-0.5	5.1	9.4	14.9	18.7	19.8	18.9	15.4	9.4	3.7	0.1		
	2%	DB	-3.7	-0.4	8.5	16.4	24.0	29.5	29.6	27.6	21.7	13.3	3.0	-1.7		
		MCWB	-4.5	-1.9	2.9	7.8	13.1	17.6	19.0	18.0	14.5	7.5	1.2	-2.4		
	5%	DB	-5.5	-2.3	5.8	13.2	20.6	27.4	27.2	25.3	18.8	10.0	0.5	-3.2		
		MCWB	-6.3	-3.7	1.2	6.2	11.1	17.0	18.1	17.3	12.6	5.9	-0.5	-3.8		
	10%	DB	-7.5	-4.1	3.6	10.6	17.4	25.0	25.0	22.7	16.1	7.5	-0.9	-5.1		
		MCWB	-8.2	-5.4	0.3	4.8	10.1	16.2	17.5	16.3	11.4	5.2	-1.8	-5.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.2	0.1	5.4	10.5	15.6	21.0	21.2	20.6	16.2	10.4	4.2	0.4		
		MCDB	-1.8	1.8	11.5	18.3	25.4	28.6	28.9	27.3	22.8	15.1	5.3	1.0		
	2%	WB	-4.4	-1.7	3.3	8.5	13.9	19.1	19.9	19.2	14.9	8.4	1.1	-2.3		
		MCDB	-3.8	-0.4	8.0	15.1	21.1	26.1	26.5	25.6	21.1	12.1	2.7	-1.8		
	5%	WB	-6.2	-3.5	1.7	6.8	12.4	18.1	19.1	18.1	13.4	6.9	-0.2	-3.8		
		MCDB	-5.5	-2.3	4.9	12.3	18.1	24.6	24.9	23.3	17.8	9.7	0.5	-3.2		
	10%	WB	-8.2	-5.5	0.5	5.3	10.9	17.0	18.2	17.0	11.9	5.2	-1.8	-5.8		
		MCDB	-7.5	-4.1	3.5	10.1	16.2	23.1	23.4	21.2	15.5	7.2	-0.9	-5.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.6	11.7	13.0	11.8	12.5	13.0	11.4	12.0	11.5	9.2	8.5	8.3		
		MCDBR	8.7	12.5	16.6	17.6	20.3	18.2	17.1	18.0	18.0	15.0	8.5	8.0		
		MCWBR	8.2	10.2	11.1	9.8	10.0	7.9	7.3	8.6	10.0	9.1	7.0	7.5		
		MCDBR	8.6	10.9	14.7	16.1	16.6	15.4	13.3	14.3	16.0	13.9	7.5	7.7		
	5% WB	MCWBR	8.1	8.9	9.7	9.4	8.9	7.3	6.3	7.3	8.7	8.8	6.3	7.3		
(40) Clear-Sky Solar Irradiance	taub		0.278	0.313	0.357	0.397	0.408	0.394	0.404	0.393	0.359	0.343	0.315	0.284		
	taud		2.046	2.009	1.953	2.041	2.197	2.355	2.354	2.354	2.391	2.272	2.053	1.979		
	Ebn at Noon		637	742	793	820	836	854	837	823	806	715	571	520		
	Edh at Noon		70	104	139	147	134	116	114	107	91	81	70	61		
(44) All-Sky Solar Radiation	RadAvg		0.97	1.99	3.32	4.25	5.00	5.92	5.28	4.40	3.00	1.65	0.84	0.63		
	RadStd		0.10	0.14	0.32	0.27	0.59	0.61	0.42	0.39	0.37	0.16	0.12	0.08		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47) Regional (14 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page